THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — SCIENCE · CLASS 9 · SET 04

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — BIOLOGY — 16 questions · 32 marks

Q.1–7 MCQs (1 mark each); Q.8–9 Assertion–Reason. For Assertion–Reason: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

1. Which cell organelle contains its own DNA and ribosomes and can therefore make some of its own proteins?

(a) Golgi apparatus (b) Mitochondria
(c) Lysosome (d) Endoplasmic reticulum

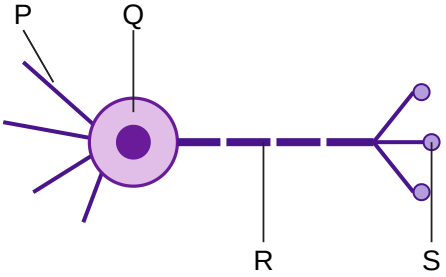
Ans: (b) Mitochondria (and plastids) have their own DNA and ribosomes.

2. A cube of potato kept in a strong sugar solution for an hour loses mass. This is because:

(a) the sugar solution is hypotonic to the potato cells
(b) the potato cells are hypertonic to the solution
(c) water moves out of the potato cells by osmosis since the solution is hypertonic
(d) sugar enters the potato cells by osmosis

Ans: (c) A hypertonic external solution draws water out of the cells.

3. In the neuron shown, the part that carries the nerve impulse away from the cell body is:



(a) P (b) Q (c) R (d) S

Ans: (c) The axon (R) carries impulses away from the cell body.

4. A muscle has spindle-shaped cells, each with a single nucleus, no striations and works involuntarily. It is:

(a) skeletal muscle (b) cardiac muscle
(c) smooth muscle (d) striated muscle

Ans: (c) These are features of smooth (unstriated) muscle.

5. Which of the following would NOT help in reducing the enhanced greenhouse effect?

(a) planting more trees (b) using solar energy
(c) burning more coal (d) using public transport

Ans: (c) Burning coal releases CO_2.

6. Which of the following is a non-flowering seed plant (gymnosperm)?

(a) Fern (b) Pine (c) Moss (d) Mango

Ans: (b) Pine bears naked seeds in cones.

7. Which of the following is NOT a method of vegetative propagation?

(a) Grafting (b) Layering (c) Pollination (d) Cutting

Ans: (c) Pollination is a step in sexual reproduction.

8. **Assertion (A):** Xylem transports water and minerals from the roots to the leaves.
Reason (R): Xylem vessels and tracheids are dead cells with lignified walls that form continuous tubes.

Ans: (a) The continuous hollow tubes explain how xylem conducts water.

9. **Assertion (A):** Insolation at the Earth's surface is always equal to the solar constant.
Reason (R): Part of the solar radiation is absorbed and reflected by the atmosphere.

Ans: (d) A is false – surface insolation is less than the solar constant. R is true.

10. Why is the cell membrane called selectively permeable?

It allows only certain substances (like water, gases, nutrients) to pass in or out [1] and prevents others, controlling the cell's internal environment. [1]

O R

What are plastids? Name two types of plastids.

Organelles found in plant cells that contain pigments / store food [1]; chloroplast, chromoplast, leucoplast (any two). [1]

11. Name the tissue that (i) stores fat in the body, (ii) forms the outer covering of the skin and (iii) carries oxygen in the blood.

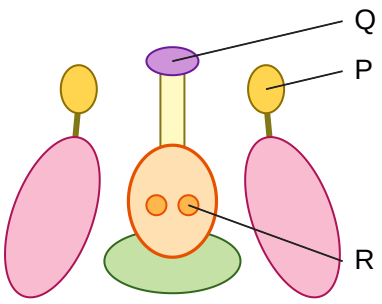
(i) Adipose tissue [1] (ii) Stratified squamous epithelium [1] (iii) Blood (red blood cells) [1]

O R

What is cartilage? Name two places where it is found in the body and state one function.

Smooth, flexible connective tissue with widely spaced cells in a firm matrix [1]; nose, ear, trachea, joints [1]; support and reduces friction at joints. [1]

12. The figure shows a flower. Name the parts P, Q and R. In which of these does fertilisation occur?



P – Anther (produces pollen), Q – Stigma, R – Ovule [½ each + function wording = 2]; fertilisation occurs in the ovule (R). [1]

13. Match the organelles with their functions by filling the table:

Organelle	Function
Lysosome	(a) _____
Vacuole	(b) _____
Plastids	(c) _____

(a) Digests worn-out cell parts / foreign material – 'suicide bag' [1] (b) Stores cell sap, water, wastes [1] (c) Contain pigments – photosynthesis (chloroplast) / storage [1]

14. Explain how chlorofluorocarbons (CFCs) deplete the ozone layer. State two harmful effects of increased ultraviolet radiation on living organisms.

CFCs rise to the stratosphere, UV breaks them to release chlorine atoms which destroy ozone molecules (catalytic chain reaction). [1½] Effects – skin cancer, cataract, damage to crops/phytoplankton (any two). [1½]

15. A botanist found three plants in a forest. Plant X has no vascular tissue and grows in a damp, shady place. Plant Y has vascular tissue but reproduces by spores and does not produce seeds. Plant Z produces naked seeds in cones and has needle-like leaves.

(a) To which group does plant X belong?

Bryophyta. [1]

(b) To which group does plant Y belong?

Pteridophyta. [1]

(c) Name the group of plant Z and give one example. State one more feature of the group.

Gymnosperms; pine/cycas/deodar [1]; no flowers or fruits, seeds are not enclosed in an ovary [1]

OR

(c) Why are bryophytes called the ‘amphibians of the plant kingdom’?

They live on land but need water for fertilisation (the male gamete swims to the egg) [2]

16. Describe the structure of an animal cell with a neat labelled diagram showing at least five parts. State the function of any two parts.

Diagram – cell membrane, cytoplasm, nucleus, mitochondria, Golgi apparatus, ER, ribosomes, lysosomes (any five labelled correctly) [3]; functions of any two [2]

OR

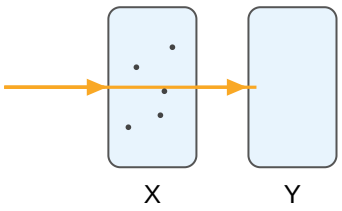
(a) Define diffusion and osmosis. (b) Distinguish between diffusion and osmosis on any three points. (c) Name the process by which a raisin swells in water.

(a) Diffusion – movement of particles from higher to lower concentration; osmosis – diffusion of water through a semi-permeable membrane [2] (b) any three: medium, membrane needed, substance moving, etc. [2] (c) Osmosis (endosmosis) [1]

SECTION B — CHEMISTRY — 11 questions · 25 marks

Q.17–19 MCQs; Q.20 Assertion–Reason (options as in Section A).

17. Light is passed through two test tubes X and Y. The path of light is visible in X but not in Y. X and Y could respectively be:



(a) copper sulphate solution and milk
(b) milk and salt solution
(c) salt solution and sugar solution
(d) sugar solution and salt solution

Ans: (b) Milk (colloid) scatters light; salt solution (true solution) does not.

18. The number of atoms present in one molecule of sulphuric acid (H_2SO_4) is:

(a) 3 (b) 4 (c) 7 (d) 6

Ans: (c) 2 + 1 + 4 = 7.

19. Which of the following is a pair of isotopes?

(a) ^{14}C and ^{14}N (b) ^{35}Cl and ^{37}Cl (c) ^{40}Ar and ^{40}Ca (d) Na and K

> **Ans: (b)** Same atomic number (17), different mass numbers.

20. **Assertion (A):** Noble gases are chemically inert.
Reason (R): Noble gases are present in the atmosphere.

> **Ans: (b)** Both true; inertness is due to a complete outer shell, not their presence in air.

21. Two beakers contain (i) chalk powder in water and (ii) salt in water. A torch beam is passed through each. In which beaker is the beam visible? How can chalk be separated from the water?

> Beaker (i) [1]; by filtration (or decantation). [1]

22. What is meant by formula unit mass? Calculate it for sodium chloride (Na = 23 u, Cl = 35.5 u).

> Sum of atomic masses of all atoms in one formula unit of an ionic compound [1]; NaCl = 23 + 35.5 = 58.5 u [1]

23. Calculate the number of protons and neutrons in $^{37}_{17}Cl$. Why do isotopes have similar chemical properties?

> Protons 17, neutrons 20 [1]; same electronic configuration/valency since the same number of electrons. [1]

24. 15 g of salt is dissolved in 85 g of water at a temperature where the solubility of salt is 36 g per 100 g of water. (i) Find the % m/m of the solution. (ii) How much more salt can be dissolved in this water to saturate it?

> (i) 15/100 × 100 = 15 % [1] (ii) Maximum in 85 g water = 36 × 85/100 = 30.6 g [1]; more = 30.6 − 15 = 15.6 g [1]

25. Explain the Tyndall effect. Give two daily-life examples.

> Scattering of a light beam by colloidal particles/suspension making its path visible [1½]; sunlight through mist/dusty room/forest canopy, torch beam in fog [1½]

O R

State three differences between a compound and a mixture.

> Composition fixed/variable; properties new/retained; separation chemical/physical; energy changes; homogeneous. [Any 3 × 1]

26. A laboratory technician has to separate the following mixtures.

Mixture	Components
1	Cream and milk
2	Different dyes in an ink
3	Ethanol (b.p. 78 °C) and water (b.p. 100 °C)
4	Common salt and ammonium chloride

(a) Name the technique suitable for mixture 1.

> Centrifugation. [1]

(b) Name the technique suitable for mixture 3.

> Distillation (fractional). [1]

(c) Name the technique for mixture 2 and state its principle.

> Chromatography [1]; different components have different solubility in the solvent / adsorption on paper. [1]

OR

(c) Name the technique for mixture 4 and state the principle.

> Sublimation [1]; ammonium chloride changes directly from solid to vapour on heating, salt does not. [1]

27. Describe an activity to show that a true solution does not scatter light while a colloid does. Write the materials, procedure, observations and conclusion.

> Materials – torch, two beakers, salt solution, milk/starch solution [1]; Procedure – pass a beam of light through each [1]; Observation – beam visible in milk, not in salt solution [1]; Conclusion – colloid particles are large enough to scatter light, solution particles are not [2]

OR

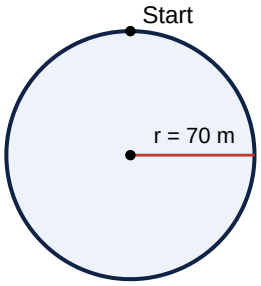
Explain how you would prepare 250 g of a 10 % (m/m) sugar solution. Calculate the masses of sugar and water required and write the steps.

> Mass of sugar = 10 % of 250 = 25 g [1]; mass of water = 225 g [1]. Steps: weigh 25 g sugar [1], add to 225 g water in a beaker [1], stir till completely dissolved [1].

SECTION C — PHYSICS — 12 questions · 23 marks

Q.28–33 MCQs; Q.34 Assertion–Reason (options as in Section A).

28. A cyclist completes one round of a circular track of radius 70 m. Taking π = 22/7, the distance covered and the displacement are respectively:



(a) 440 m, 0 (b) 0, 440 m (c) 220 m, 140 m (d) 440 m, 140 m

> **Ans: (a)** Distance = 2πr = 440 m; the final position equals the starting position, so displacement = 0.

29. A body is thrown vertically upwards with a velocity of 20 m s^{-1} (g = 10 m s^{-2}). The maximum height reached is:

(a) 10 m (b) 20 m (c) 40 m (d) 200 m

> **Ans: (b)** $h = v^2/2g = 400/20 = 20$ m.

30. A 5 kg block on a smooth surface is acted on by a constant force of 10 N for 3 s, starting from rest. Its velocity after 3 s is:

(a) 2 m s^{-1} (b) 6 m s^{-1} (c) 15 m s^{-1} (d) 30 m s^{-1}

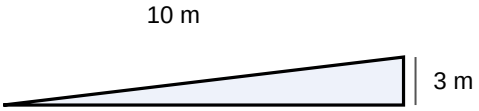
> **Ans: (b)** a = 2 m s^{-2}; v = at = 6 m s^{-1}.

31. A gun of mass 4 kg fires a bullet of mass 50 g with a velocity of 200 m s^{-1}. The recoil velocity of the gun is:

(a) 0.25 m s^{-1} (b) 2.5 m s^{-1} (c) 25 m s^{-1} (d) 800 m s^{-1}

> **Ans: (b)** 0.05 × 200 = 4 × v → v = 2.5 m s^{-1} (opposite direction).

32. The inclined plane shown is 10 m long and 3 m high. Its ideal mechanical advantage is:



(a) 3 (b) 3.33 (c) 0.3 (d) 7

Ans: (b) MA = length/height = 10/3 ≈ 3.33.

33. The time period of a sound wave of frequency 50 Hz is:

(a) 0.2 s (b) 0.02 s (c) 50 s (d) 0.5 s

Ans: (b) T = 1/f = 1/50 = 0.02 s.

34. **Assertion (A):** A body moving with constant velocity has zero acceleration.
Reason (R): Acceleration is the rate of change of speed only, not of direction.

Ans: (c) A is true. R is false – acceleration is the rate of change of velocity (speed and/or direction).

35. A cricket ball of mass 0.15 kg moving at 30 m s^{-1} is stopped by a fielder in 0.03 s. Calculate (i) the change in momentum and (ii) the average force exerted by the ball on the fielder's hands.

(i) $\Delta p = 0.15 \times 30 = 4.5$ kg m s^{-1} [1] (ii) $F = 4.5/0.03 = 150$ N [1]

36. The speed of a body is doubled. How does its kinetic energy change? Give a reason.

It becomes four times [1]; KE ∝ v^2. [1]

OR

Define power. Write its SI unit and define it.

Power – rate of doing work [½]; SI unit watt [½]; 1 W = 1 J of work done in 1 s. [1]

37. A train 150 m long crosses a pole in 10 s moving at uniform speed. Find (i) the speed of the train and (ii) the time taken to cross a bridge 300 m long.

(i) v = 150/10 = 15 m s^{-1} [1½] (ii) distance = 150 + 300 = 450 m; t = 450/15 = 30 s [1½]

38. Priya, a 50 kg student, climbs a flight of stairs 6 m high in 20 s in her school building.

(a) Calculate the gain in her potential energy (g = 10 m s^{-2}).

PE = mgh = 50×10×6 = 3000 J. [1]

(b) Find the power she develops.

P = 3000/20 = 150 W. [1]

(c) Explain why no work is done by gravity on Priya when she walks on a level corridor.

Gravity acts vertically downwards while displacement is horizontal – perpendicular to the force [1]; so W = Fs cos 90° = 0. [1]

OR

(c) State the energy conversion when a ball falls freely from a height.

Potential energy decreases and is converted into kinetic energy [1]; total mechanical energy remains constant (if air resistance is neglected). [1]

39. A train starting from rest accelerates at 0.5 m s^{-2} for 1 minute, runs at constant speed for 2 minutes and then decelerates uniformly to rest in 30 s. Find (i) the maximum speed, (ii) the distance in the first minute, (iii) the distance at constant speed, (iv) the retardation and (v) the total distance covered.

(i) $v = 0.5 \times 60 = 30$ m s^{-1} [1] (ii) ½×0.5×3600 = 900 m [1] (iii) 30 × 120 = 3600 m [1] (iv) 30/30 = 1 m s^{-2} [1] (v) 900 + 3600 + ½×30×30 (450) = 4950 m [1]

O R

Derive the relation $v^2 - u^2 = 2as$ for a body moving with uniform acceleration, using the velocity–time graph.

Graph with u at t = 0 to v at t [1]. s = area = (u+v)/2 × t [1½]; a = (v−u)/t → t = (v−u)/a [1½]; s = (v+u)(v−u)/2a → $2as = v^2 - u^2$. [1]